

Connor Porter

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Summary

Mathematics graduate with strong analytical, mathematical, programming, and teaching skills. Currently seeking PhD opportunities, with particular interests in algebraic topology, Lie theory, category theory, and computational mathematics. Research experience includes work on classifying spaces, simplicial homology and cohomology, alongside coursework in Lie algebras. Proficient in LaTeX, Python, and mathematical computing tools, with practical experience gained through academic projects and professional experience.

Technical Skills

- | | | |
|----------|--------------|---------------------------|
| • LaTeX | • Pandas | • NetworkX |
| • Python | • NumPy | • SageMath |
| • Git | • Matplotlib | • Optimisation Algorithms |

Education

Cardiff University

Cardiff, Wales

Master of Science in Mathematics

23/09/2024-08/09/2025

- **Classification:** Distinction | 85%
- **Dissertation:** Categories and Classifying Spaces (80%)
- **Relevant Modules:**
 - Lie Algebras (92%)
 - Reading Module – Operator Theory and C* Algebras (85%)
 - Mathematical Biology (85%)

Cardiff University

Cardiff, Wales

Bachelor of Science in Mathematics

04/10/2021-17/07/2024

- **Classification:** First Class Honours Degree | 75%
- **Scholarship Award:** Cardiff University Scholarship Award: Awarded £3,000
- **Relevant Modules:**
 - Algebraic Topology (79%)
 - Mathematical Investigations with Python (68%)
 - Algebra II: Rings (73%)
 - Algebra I: Groups (94%)

Relevant Work Experience

Office for National Statistics

Newport/London, Wales/England

Analytical Internship

03/07/2023-03/10/2023

- Used Python to perform automated quality checks on new data releases and integrate validated data into the pipeline.

- Contributed to the development of the Joint Data Access Centre (JDAC), working with Python and Git to process and manage data.
- Awarded a £50 gift voucher in recognition of outstanding performance during the internship.

Self Employed

Mathematics Tutor

Abercarn, Wales

07/10/2024-Present

- Established and operate an independent mathematics tutoring business, building a regular client base and improving my ability to explain complex ideas simply.
- Provide face to face tutoring for GCSE and A-Level students, enhancing their understanding of mathematical concepts.

Relevant Projects

Categories and Classifying Spaces | 80%

Cardiff University

Investigated a relationship within simplicial homology, proving that the simplicial homology groups of a group are isomorphic to the simplicial homology groups of its classifying space. Used the nerve construction to associate a simplicial set to a group viewed as a category and, following Hatcher (2002), proved this has a natural Delta-complex structure. Verified the theory through computations such as the classifying space of the cyclic group of order n . Also extended the study to cohomology and applied the Universal Coefficient Theorem to relate the cohomology and homology groups of the classifying space.

Lie Groups and their relation to Lie Algebras | 90%

Cardiff University

Explored the relationship between Lie groups and Lie algebras, focusing on tangent spaces, exponential maps, and Lie algebra isomorphisms. Studied $SO(3)$ and $SU(2)$ as concrete examples, demonstrating through explicit Lie bracket calculations that non-isomorphic Lie groups can have isomorphic Lie algebras. Discussed applications of Lie groups in representation theory, differential geometry, and physics.

Lie Algebras Coursework | 92%

Cardiff University

Studied the structure of the symplectic Lie algebra $\mathfrak{sp}(2n)$, determining its dimension and constructing a Cartan subalgebra. Calculated the corresponding root system, coroots, and Killing form through explicit matrix calculations. Constructed an $\mathfrak{sl}(2)$ -triple within $\mathfrak{sp}(2n)$ and verified the required commutation relations directly.

Shortest Path Between Doctors and Patients | 72%

Cardiff University

A Python group project finding efficient routes between doctors and patients. Used local search and the Great Deluge algorithm to iteratively improve randomly generated routes while considering constraints on the number of patients assigned to each doctor. Also used GPS data and the Haversine formula to apply the algorithms to real-world geographical locations. Used NetworkX and matplotlib to visualise routes and analyse the convergence and performance of the optimisation algorithms.

Statistical Distribution Sampling Library | 100%

Cardiff University

A Python library developed as part of a group project for generating random samples from a range of statistical distributions: the normal, exponential, Poisson, and continuous and discrete uniform distributions. It allows users to input distribution parameters and random

seeds and can also be used to generate probability density and cumulative distribution plots using matplotlib. As an example, we modelled the throws of a darts player using independent normal distributions to represent their horizontal and vertical accuracy.

Volunteer Teaching Experience

Royal Air Force Air Cadets

Blaina, Wales

Training Officer of 275 Nantyglo and Blaina Squadron

18/06/2024-Present

- Create and deliver monthly training programmes covering a range of theoretical and practical subjects including Principles of Flight, Flying Operations, fieldcraft, weapons training, and STEM.
- Oversee weapons training and marksmanship activities across six local squadrons, supporting approximately 100-120 cadets.
- Develop and plan lessons and activities to explain difficult concepts to cadets of different ages and abilities.

Extracurricular University Experience

Cardiff University Amateur Boxing Club

Cardiff, Wales

President

23/09/2024-08/09/2025

- Introduced a new weekly women-only training session to increase accessibility and participation within the club.
- Liaised with the club's Wellbeing Officer and Student Union to promote an inclusive and supportive training environment for all members.
- Organised and delivered a successful boxing home show, raising over £600 to replace and upgrade the club's ageing boxing equipment.

Treasurer and Vice Captain

18/09/2023-23/09/2024